BASE CHANGE PRESERVES THE TOR AMPLITUDE INTERVAL

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Characterisation of tor amplitude through a complex of flat modules concentrated in the interval. Lemma \ref{lemma-tor-amplitude}, cited at proof lines 17157-17159 and 17165-17167. Provides the flat replacement and the criterion applied over the new ring.
- Base change of a flat module is flat. Algebra, Lemma \ref{algebra-lemma-flat-base-change}, cited at proof lines 17163-17164. Makes the base changed complex consist of flat modules.

Source reference. Original lemma 066L: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7940–7940. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0654: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7888–7888. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00HI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 644–644. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let $A \to B$ be a ring map. Let $a, b \in \mathbf{Z}$. Let $K^\bullet$ be a complex of A-modules with tor amplitude in $[a, b]$. Then $K^\bullet \otimes^{\mathbf{L}}_A B$ as a complex of B-modules has tor amplitude in $[a, b]$.*

Proof. By Lemma 0654 we can find a quasi-isomorphism $E^\bullet \to K^\bullet$ where $E^\bullet$ is a complex of flat A-modules with $E^i = 0$ for $i \notin [a, b]$. Then $E^\bullet \otimes_A B$ computes $K^\bullet \otimes^{\mathbf{L}}_A B$ by construction and each $E^i \otimes_A B$ is a flat B-module by Algebra, Lemma 00HI. Hence we conclude by Lemma 0654. □